

CS & IT ENGINEERING



Data Structure & Programming

Trees and Graphs

Lecture No.- 01



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Topics to be Covered



- H/W Q's Solving

- Trees and Graphs

- Tree Fundamentals

- Degree

- height

- depth, Level

- Terminology

- Types of Binary Trees



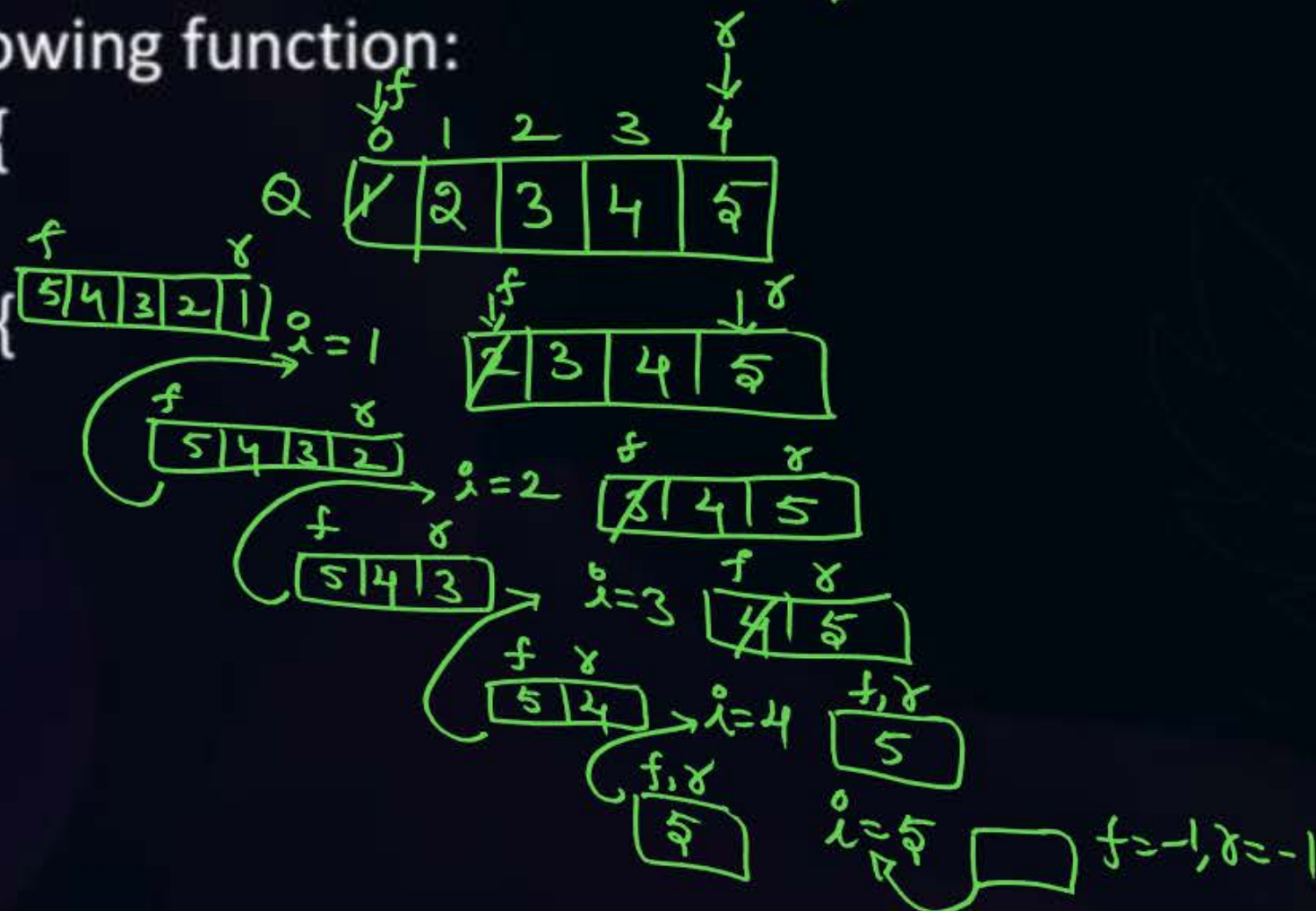


#Q. Suppose you are given an implementation of a queue of integers. The operations that can be performed on the queue are:

- i. isEmpty (Q) — returns true if the queue is empty, false otherwise.
- ii. delete (Q) — deletes the element at the front of the queue and returns its value.
- iii. insert (Q, i) — inserts the integer i at the rear of the queue.

Consider the following function:

```
void f (queue Q) {  
    int i ;  
    if (!isEmpty(Q)) {  
        i = delete(Q);  
        f(Q);  
        insert(Q, i);  
    }  
}
```





What operation is performed by the above function f ?

- (a) Leaves the queue Q unchanged
- ☒ (b) Reverses the order of the elements in the queue Q
- (c) Deletes the element at the front of the queue Q and inserts it at the rear keeping the other elements in the same order
- (d) Empties the queue Q



Topic : PYQ Practice – 1 – GATE 2006

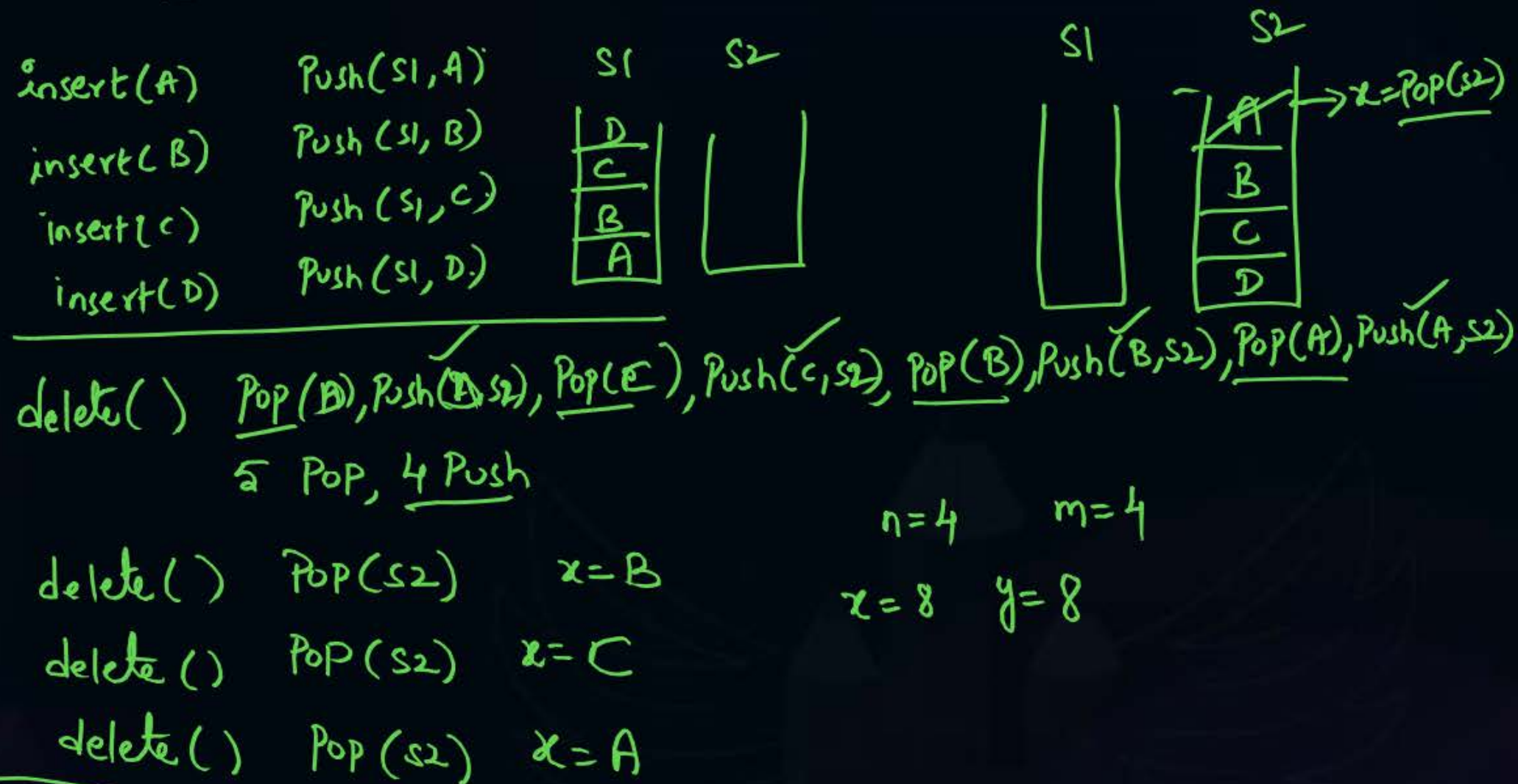


An implementation of a queue Q, using two stacks S1 and S2 is given below

```
void insert (Q, x) {
    push (S1, x);
}

void delete (Q) {
    if (stack-empty (S2)) then
        if (stack-empty (S1)) then {
            print ("Q is empty");
            return;
        }
        else while (! (stack-empty (S1))) {
            x = pop (S1);
            push (S2, x);
        }
        x = pop (S2);
    }
}
```

Handwritten notes:
 $n \text{ insert}() \Rightarrow n \text{ push}()$
 $m \text{ delete}()$
shift elements from S1 to S2
Pop (n-m) elements from S1, and Push to S2, Pop from S1



$$n = 4 \quad m = 4$$

$$x = 8 \quad y = 8$$

Let n insert and m ($\leq n$) delete operations be performed in an arbitrary order on an empty queue Q. Let x and y be the number of push and pop operations performed respectively in the process. Which one of the following is true for all m and n?



Topic : PYQ Practice - 1

Insert (A, B, C, D)

Delete, Delete, Delete, Delete



- A. $n + m \leq x < 2n$ and $2m \leq y \leq n + m$ True
 B. $n + m \leq x \leq 2n$ and $2m \leq y \leq 2n$ True
 C. $2m \leq x \leq 2n$ and $2m \leq y \leq n + m$ True
 D. $2m \leq x < 2n$ and $2m \leq y \leq 2n$

$n = \text{insertions}$

$m = \text{deletions}$

$n = 4, m = 4,$

$x = 8, y = 8$

$x = \text{Push}$

$y = \text{Pop}$

- A) $8 \leq 8 < 8$ and $8 \leq 8 \leq 8$
 B) $8 \leq 8 \leq 8$ and $8 \leq 8 \leq 8$
 C) $8 \leq 8 \leq 8$ and $8 \leq 8 \leq 8$
 D) $8 \leq 8 < 8$ and $8 \leq 8 \leq 8$

n $1 < 2r$
 $1 \text{ insert}() \Rightarrow \text{Push}()$
 $1 \text{ delete}() \Rightarrow \text{Pop}(s1), \text{Push}(s2), \text{Pop}(s2) \Rightarrow 2 \text{ Pop}(), \text{Push}()$

$n = 4, m = 3$

$\downarrow$
4 Push

1st deletion: 4 Push(s2), 4 Pop(s1), 1 Pop(s2)

2nd deletion: 1 Pop(s2)

3rd deletion: 1 Pop(s2)

7 Pop()

$x = 8, y = 7$

- A) $7 \leq 8 < 8$ and $6 \leq 7 \leq 7$
 B) $7 \leq 8 \leq 8$ and $6 \leq 7 \leq 8$
 C) $6 \leq 8 \leq 8$ and $6 \leq 7 \leq 7$
 D) $6 \leq 8 < 8$ and $6 \leq 7 \leq 8$



Topic : PYQ Practice - 1



$$4 \text{ insertions} \Rightarrow n=4 \Rightarrow x=4$$

$$0 \text{ deletions} \Rightarrow m=0 \Rightarrow y=0$$

$$n+m \leq x \leq 2n, \quad 2m \leq y \leq n+m$$

$$A) \Rightarrow 4 \leq 4 \leq 8 \quad \& \quad 0 \leq 0 \leq 4$$

$$B) \quad 4 \leq 4 \leq 8 \quad \& \quad 0 \leq 0 \leq 8$$



Trees

- Non-Linear Data structure
- Tree is also known as Hierarchical Data structure
- Set of vertices/nodes/elements and edges/links with an open circuit / No cycle in it.

$$T = \{V, E\}$$

- Tree is also termed as Acyclic Graph. [Every Tree is a Graph but not vice-versa]
- 2 Types of Nodes :
 - ① Internal nodes / Non-leaf nodes / Non-Terminal nodes.
 - Nodes with descending edge to other node(s)
 - ② External nodes / Leaf nodes / Terminal nodes.
 - A Node without falling (or) descending edge to other node.

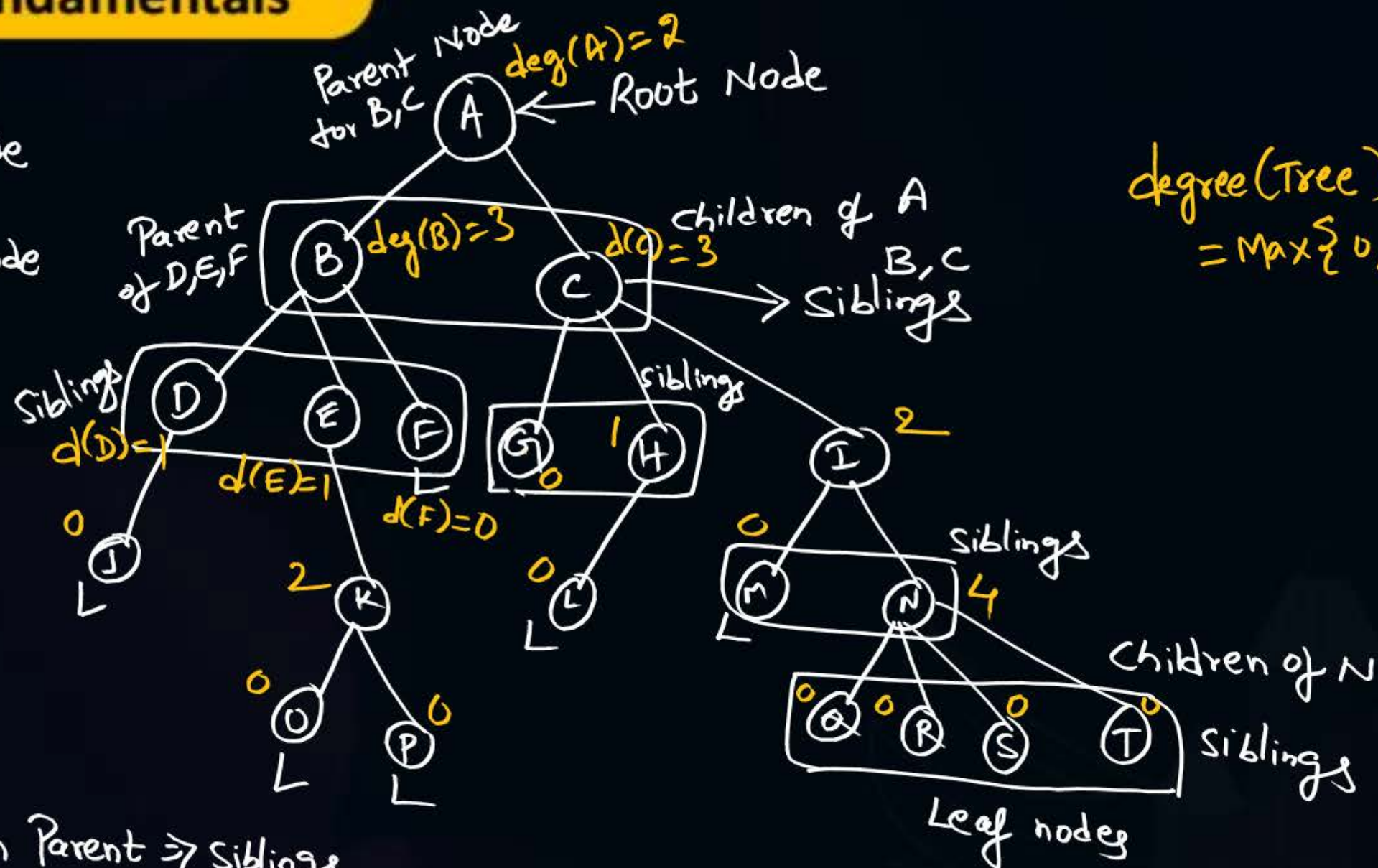


Topic : Tree Fundamentals



$\bigcirc \Rightarrow$ Unlabelled Node

$\textcircled{A} \mid \textcircled{+} \mid \textcircled{7} \Rightarrow$ Labelled Node



$$\text{degree(Tree)} = \max\{0, 1, 2, 3, 4\} = 4$$

— Nodes with Common Parent $\Rightarrow$ Siblings

$\text{Degree(Node)} = \text{Number of children of respective node}$

$\text{Degree(Leaf Nodes)} = 0$

$\text{Degree(Tree)} = \max\{\text{degrees of All nodes}\}$



Topic : Tree Fundamentals

Level of Tree $\equiv$ Depth of a Tree $\equiv$ Height of a Tree



Level, Depth, Height (Any numbering starts from zero by default).

Root Node is at Level 0.

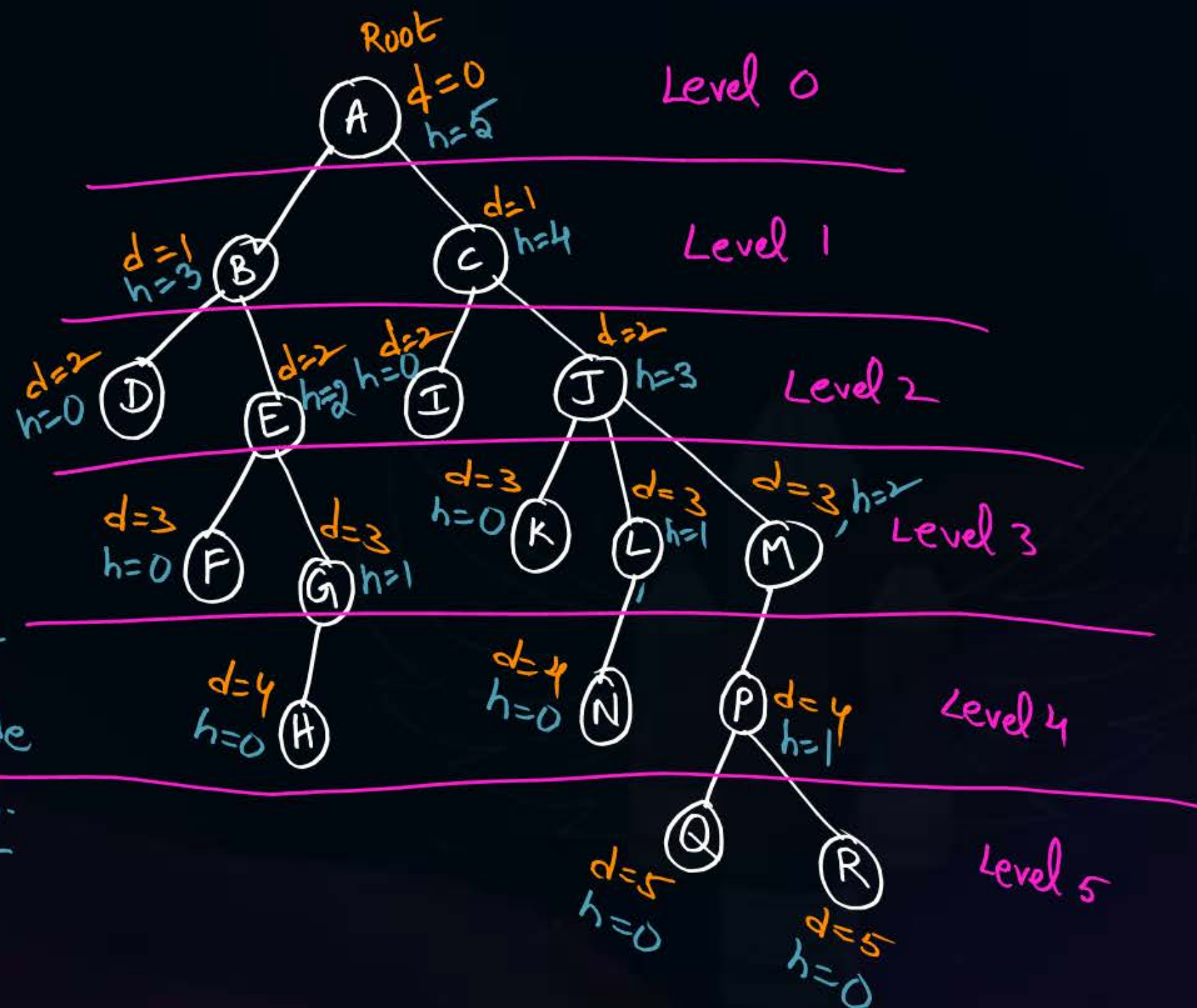
Level of a Node $\equiv$ Depth of a Node.

Depth of a Node: The Number of Edges from Root Node to respective Node.

Height of a Node: The Maximum Number of Edges from Leaf node to respective Node (OR)

The Number of Edges from Leaf node to respective node in the longest Path.

$\therefore \text{height(Leaf)} = 0$



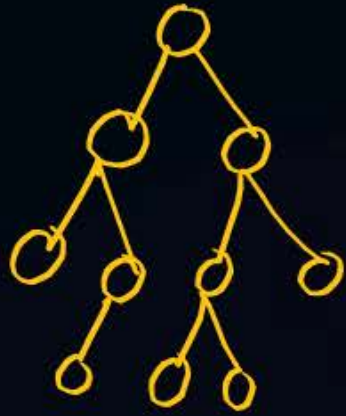


Topic : Tree Fundamentals



Binary Tree : A Tree in which degree of each node $\leq 2 \Rightarrow 0$ (or) 1 (or) 2

Ex:





Types of Binary Trees

- ① Full Binary Tree
- ② Complete Binary Tree
- ③ Perfect Binary Tree
- ④ Skewed Binary Tree
- ⑤ Degenerated Binary Tree
- ⑥ Binary Search Tree
- ⑦ Binary Heap
- ⑧ AVL Tree - - - To be discussed ...



2 mins Summary



- Tree Fundamentals

- Basic Terminology

- Types of Binary Trees

To be contd... 

THANK - YOU